URL PHISHING DETECTION

Think Before You Click!

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PLAN

1 Introduction

2 Problem Statement

3 Project Overview

4 Methodology

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6 Deployment

7 Conclusion



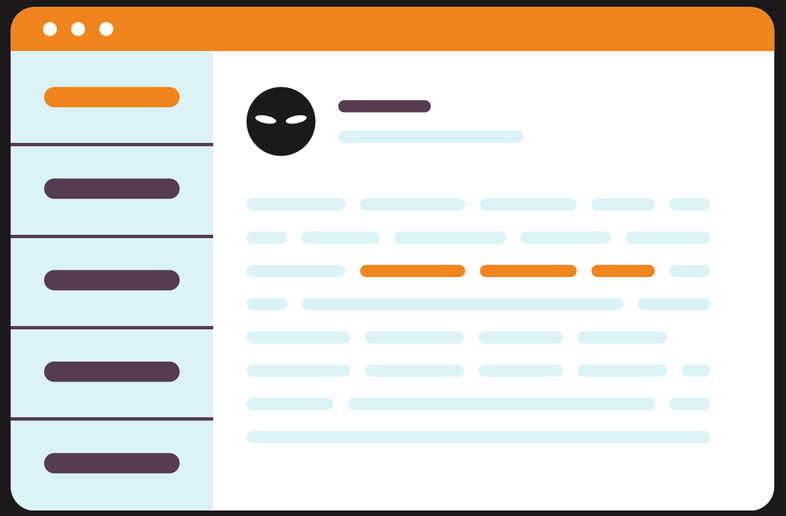
THINK BEFORE YOU CLICK!

INTRODUCTION

Don't share your personal information online!

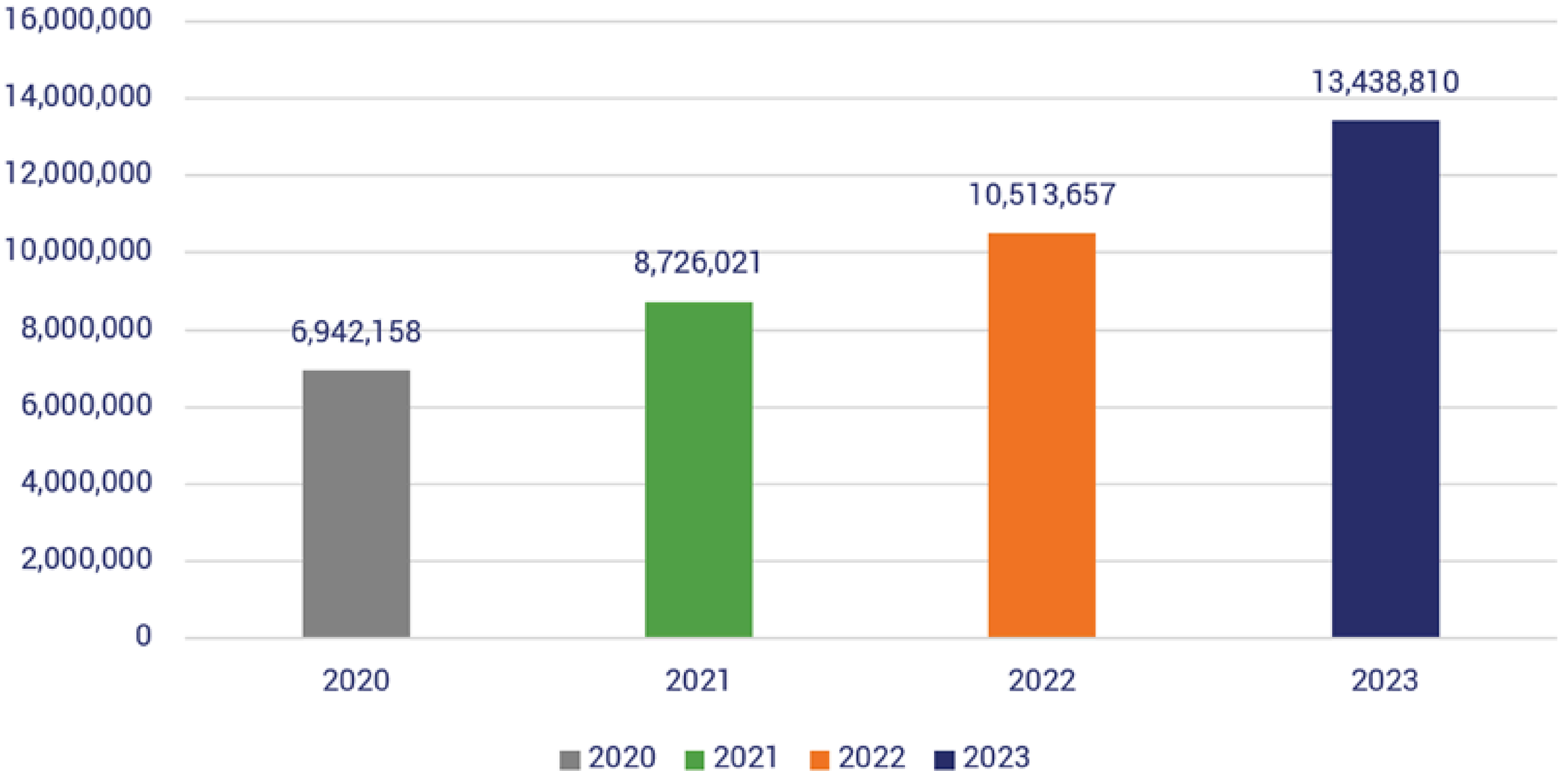
WHAT IS PHISHING?

- Phishing is a type of cyberattack aimed at stealing sensitive information like passwords or credit card numbers.
- Attackers impersonate trustworthy entities to deceive victims via an URL
- A phishing URL is a fraudulent web address that mimics legitimate websites.



Think of an email or message you received that asked for personal information. What made it suspicious?

Increasing Phishing Activity Since 2020





PHISHING URL DETECTION

- Detects deceptive web addresses that mimic legitimate sites to trick users.
- Uses algorithms and heuristics to analyze URLs for suspicious traits.

OUR PROJECT

- Focuses on building a robust system to detect phishing URLs using machine learning.

- Integrates ML models to improve detection efficiency and accuracy.





THINK BEFORE YOU CLICK!

PROBLEM STATEMENT

A fake link can sink your ship!



TRADITIONAL PHISHING DETECTION TOOLS

- Traditional methods like blacklists, heuristics, and content checks are insufficient against modern phishing attacks.

- Machine learning is now essential for strengthening cybersecurity defenses as it provides fast detection and proactive strategies



THINK BEFORE YOU CLICK!

PROJECT OVERVIEW

Hover before you discover!

OUR PROJECT

Aims to build an AI-based model for accurate phishing website detection to boost cybersecurity following 3 steps :

- **Data Collection And Preprocessing**
- **Model Training**
- **Integration and Development**



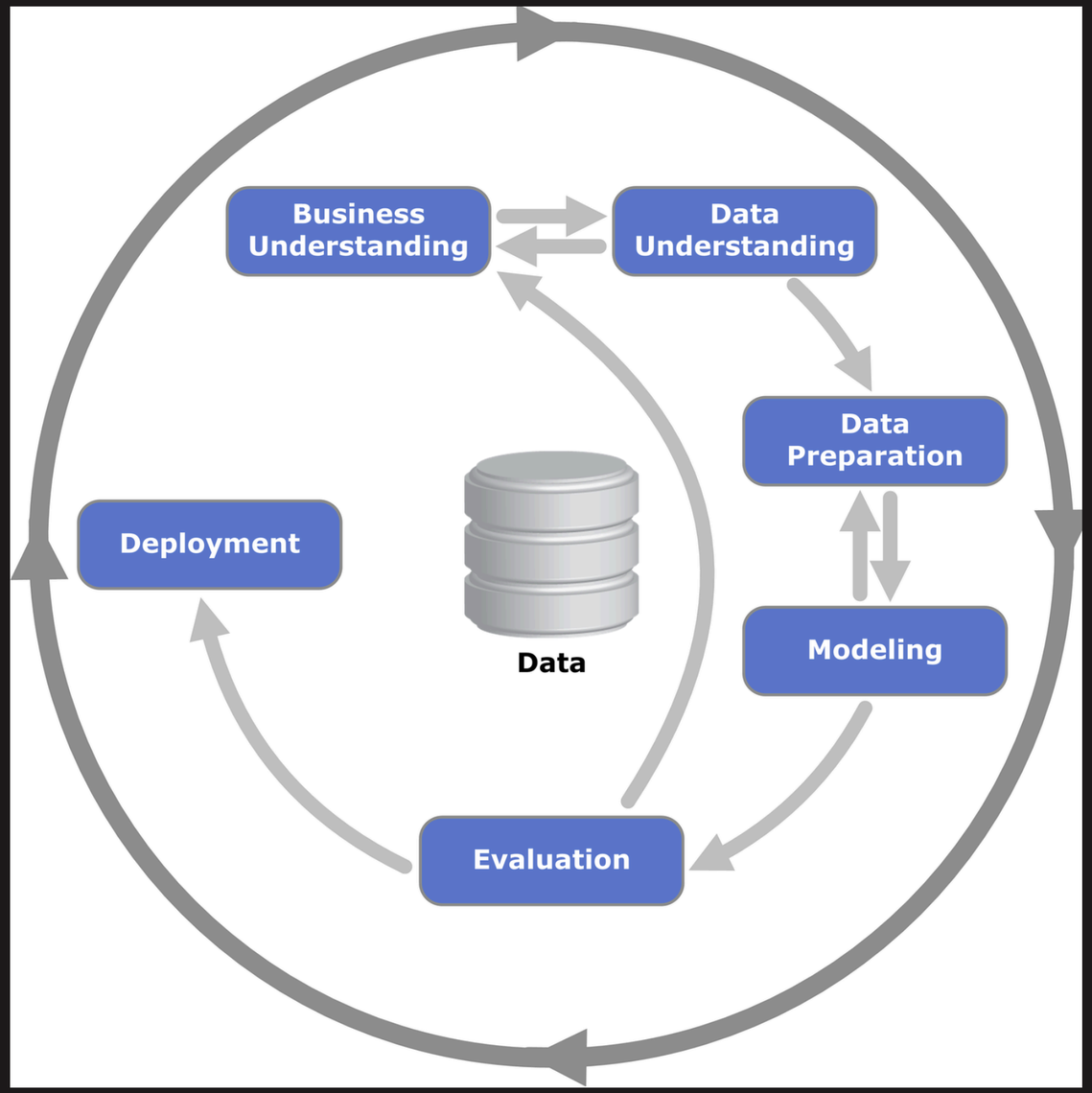


THINK BEFORE YOU CLICK!

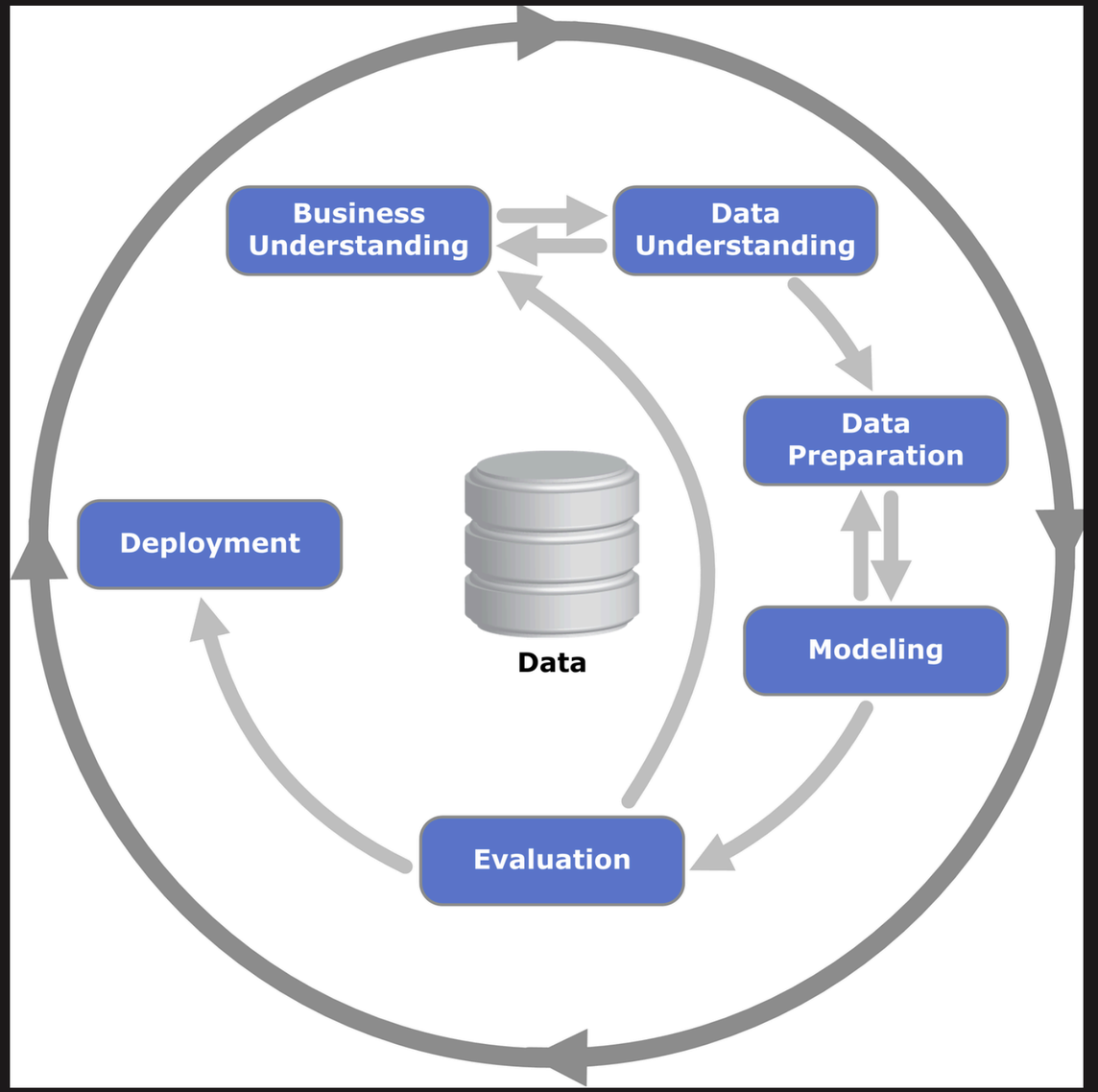
METHODOLOGY

A short URL might hide a long regret!

CRISP-DM METHODOLOGY



1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling



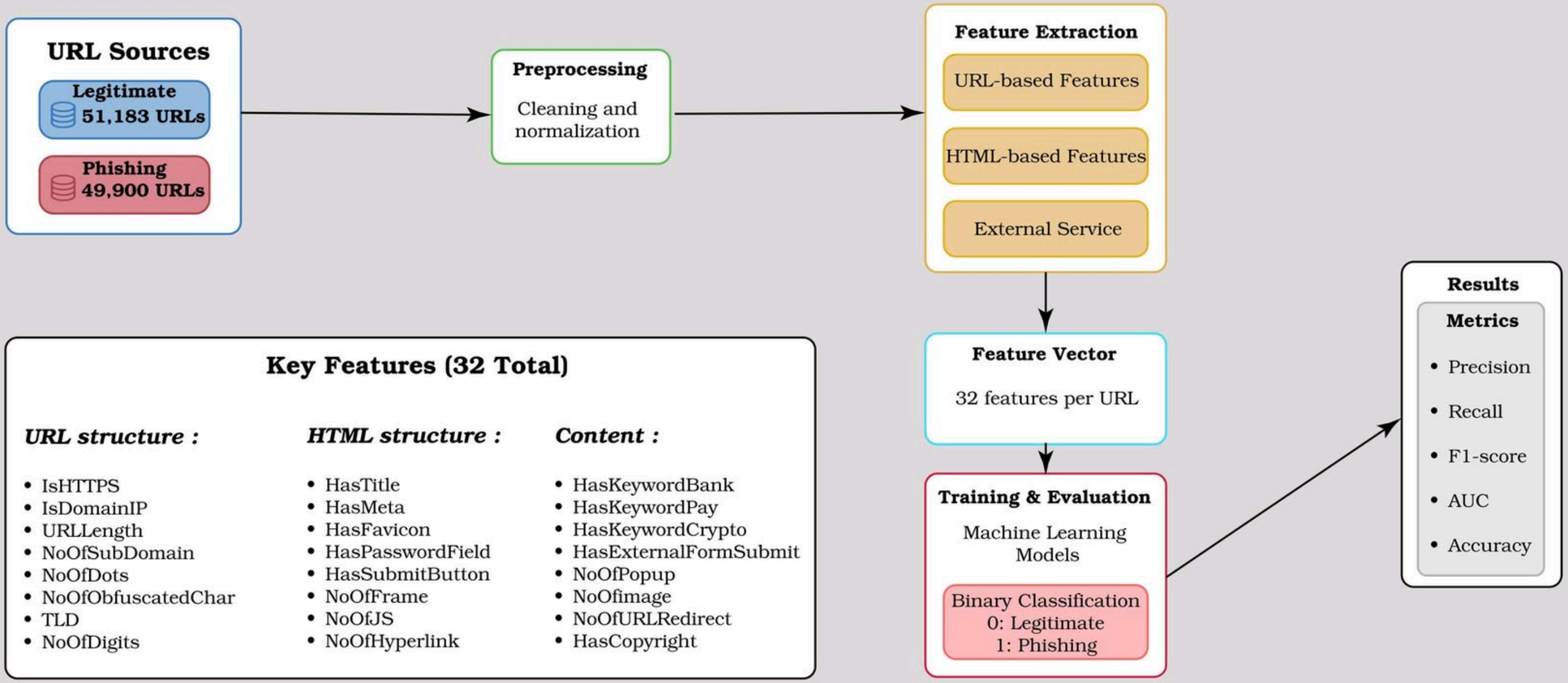


DATASETS EXPLORATION

- Explored multiple datasets sourced mainly from Kaggle and GitHub.
- Datasets contain both phishing and legitimate URLs for balanced learning.
- The variety ensures better model generalization unseen URLs.

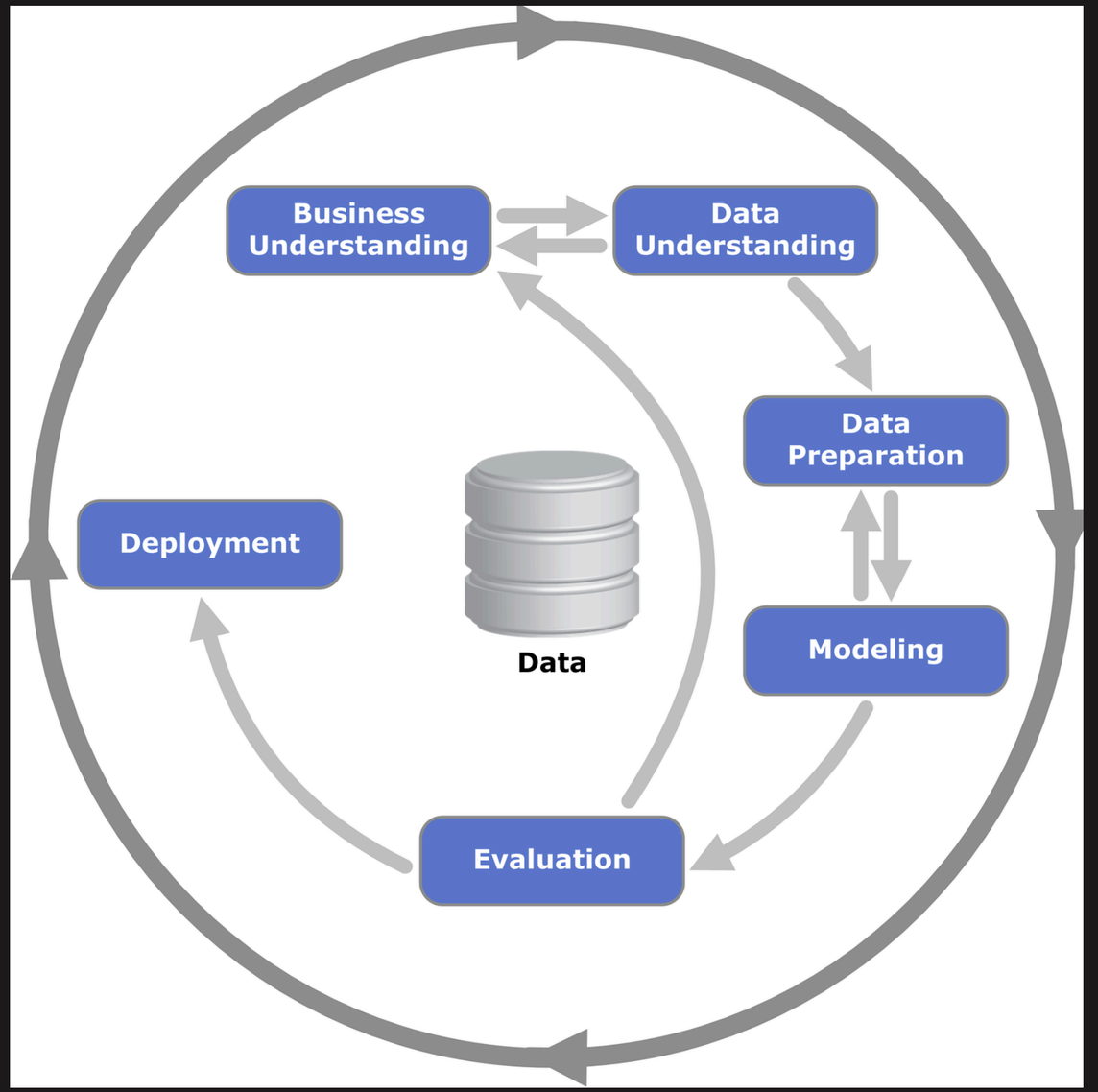
DATASET SELECTION

PhishFOE Phishing Detection System Architecture



PhishFOE Dataset: 101,083 Total URLs (51,183 Legitimate, 49,900 Phishing)

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling





01 DATA CLEANING

- **Data cleaning** removes inaccurate and incomplete entries to enhance dataset quality and reliability.



02 DATA SCALING

- **Data scaling** standardizes feature values to ensure consistency across variables, enabling more accurate model learning.



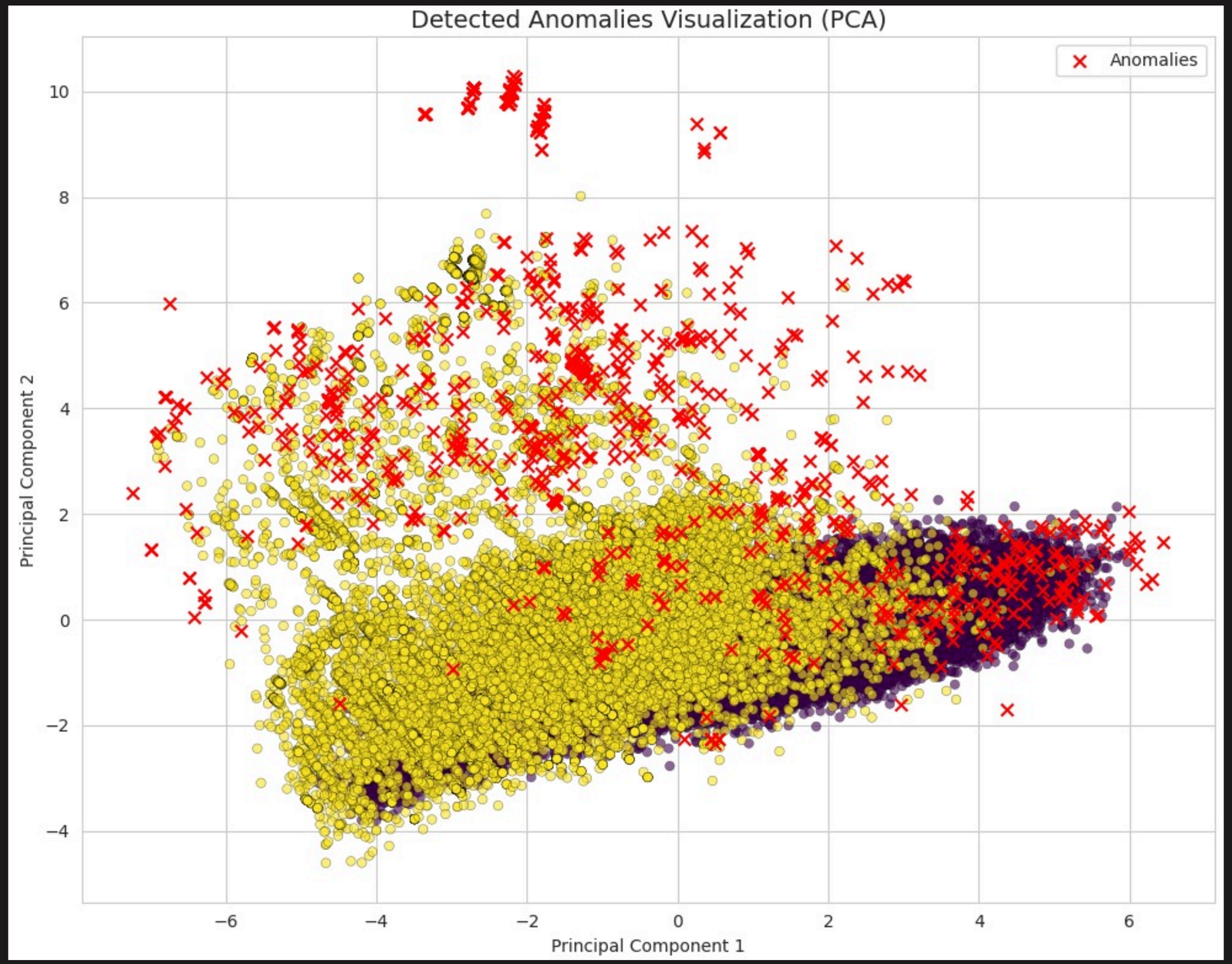
03 ANOMALY DETECTION

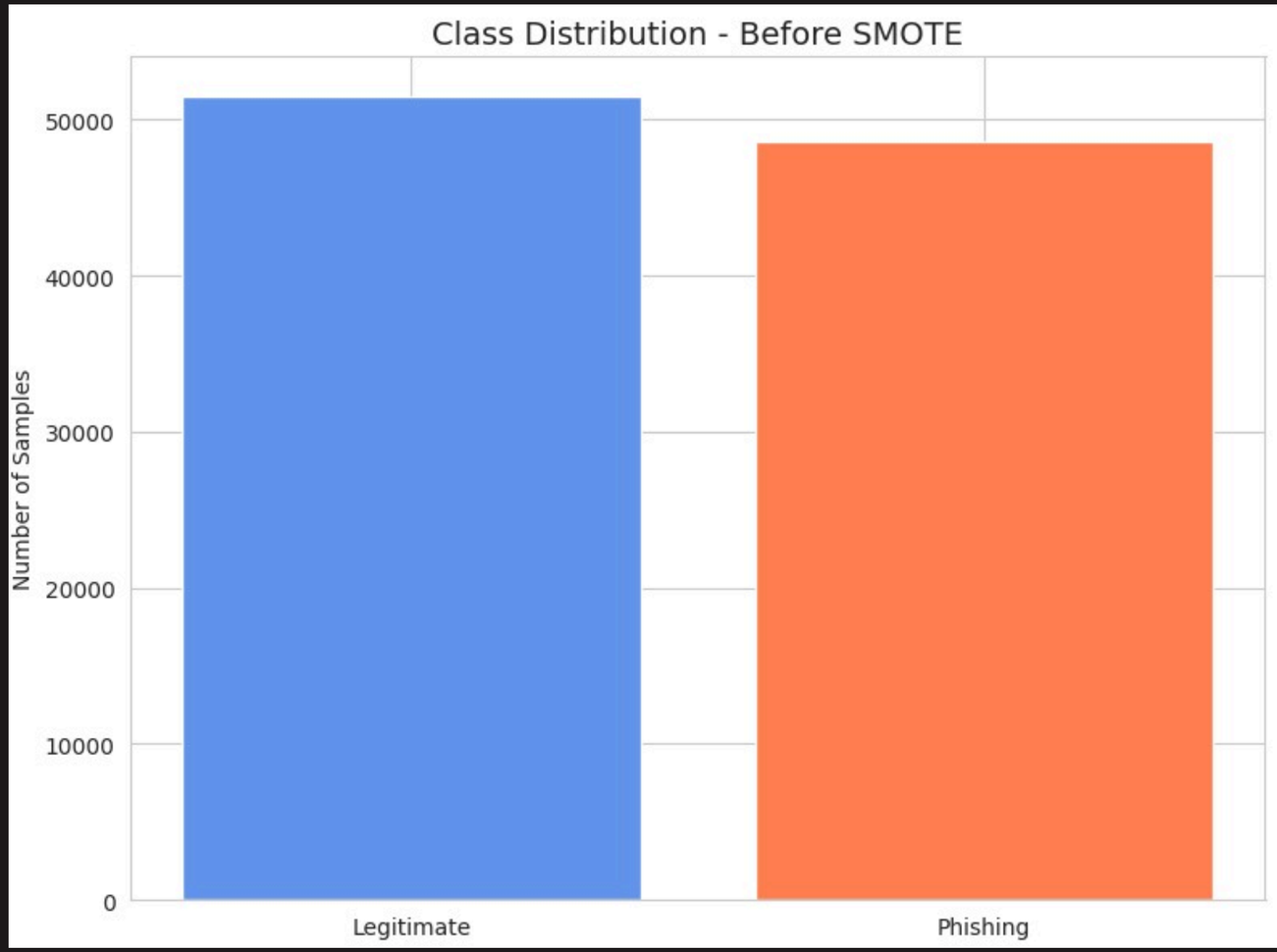
- **Anomaly detection** was performed using **Isolation Forest** to identify and remove outliers.



04 DATA BALANCING

- **Data balancing** was achieved using **SMOTE** and random **undersampling** techniques.

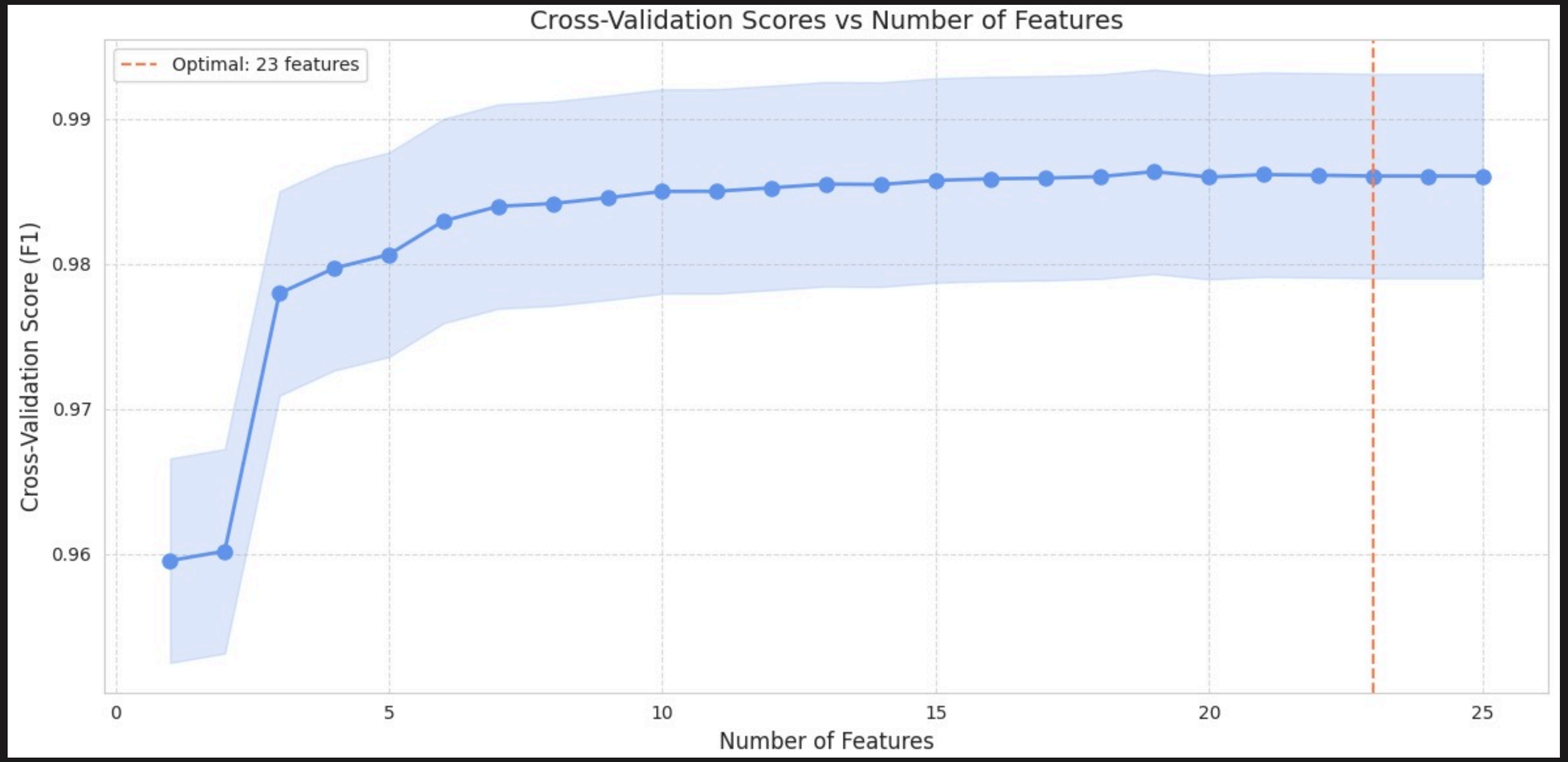

Detected Anomalies Visualization (PCA)
Anomalies
Principal Component 2
Principal Component 1
10
8
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6


Class Distribution - Before SMOTE
Number of Samples
0
10000
20000
30000
40000
50000
Legitimate
Phishing

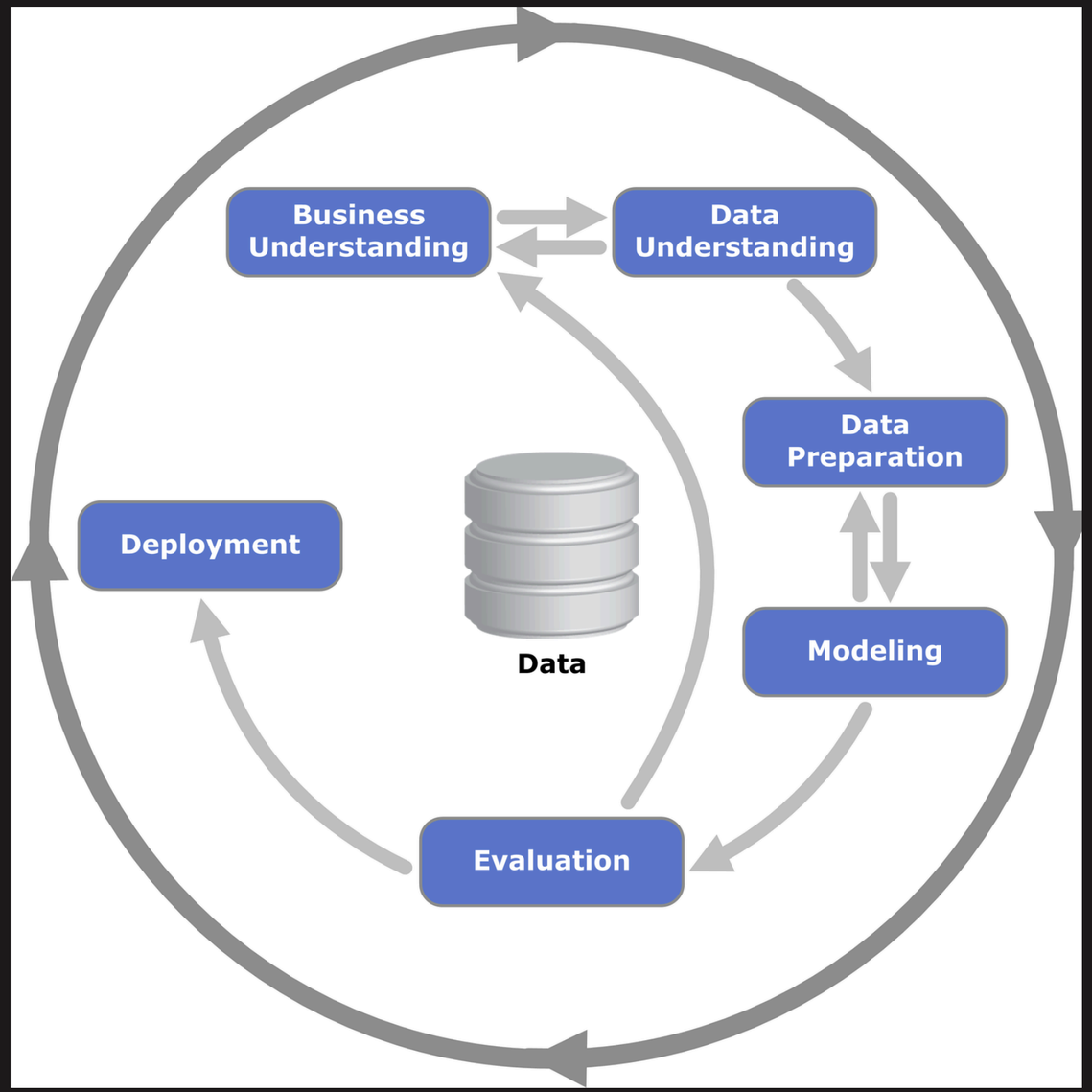

Class Distribution - After SMOTE
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0
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50000
Legitimate
Phishing

05 FEATURE SELECTION

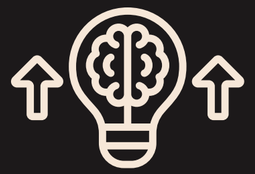
- **Feature selection** was performed using **RFE-CV** and the **f_classif** method to retain the most informative features.



1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling

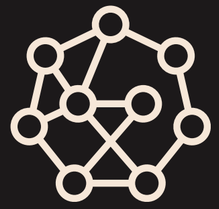


TESTED MODELS



LIGHTGBM (BOOSTING)

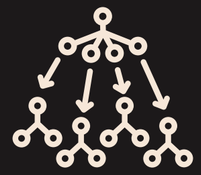
- **LightGBM** is a fast and scalable gradient boosting framework.
- It efficiently handles large datasets.
- Uses techniques like leaf-wise tree growth.



CNN (DEEP LEARNING)

- **CNN (Convolutional Neural Network**) is a deep learning model.
- CNNs automatically extract hierarchical features from structured or sequential input data.
- CNNs are well-suited for learning from URL-based phishing datasets.

TESTED MODELS



RANDOM FOREST (BAGGING)

- **Random Forest** is an ensemble learning algorithm.
- Builds multiple trees on random data subsets and combines their outputs.
- This approach improves accuracy in classification tasks.



MODEL COMBINATION LIGHTGBM , CNN (STACKING)

- Combining CNN and LightGBM creates a hybrid model.
- CNN automatically extracts deep, high-level features from data.
- LightGBM efficiently handles classification tasks.



THINK BEFORE YOU CLICK!

RESULTS AND EVALUATION

One careless click, and your info’s picked!

BENCHMARKING AGAINST PREVIOUS STUDIES

Table 3.3: Extended Comparison of Machine Learning Models for Phishing Detection Across Different Datasets

Study	Feature Selection	#Feat.	Models	Dataset	Accuracy / F1 Score
		UCI Dataset			
Liyana et al. [27] (2022)	IG, TreeSHAP	15	XGBoost	UCI	91.05% / 91.5%
Moedjahedy et al. [23] (2021)	RFE + Corr.	10	AdaBoost	UCI	94.51% / 94.2%
Al-Mohammed et al. [7] (2024)	Mutual Info	25	LightGBM	UCI	95.12% / 94.7%
Our Work	RFE-CV	23	LightGBM	UCI	**97.07% / 97.07%**
Our Work	RFE-CV	23	XGBoost	UCI	**97.03% / 96.98%**
Our Work	RFE-CV	23	CNN	UCI	**96.50% / 96.50%**
		Phishing_Legitimate_full_codes Dataset			
Ahmad et al. (2024)	Correlation Heatmap	30	RF, SVM	PLFC	97.2% / 97.1%
Khan et al. (2024)	PCA, Chi-Square	25	CNN, RNN, RF	PLFC	97.8% / 97.6%
Our Work	RFE-CV	23	LightGBM	PLFC	**99.02% / 98.89%**
Our Work	RFE-CV	23	Random Forest	PLFC	**98.70% / 98.85%**
		Phishing_Detection_Dataset			
Almsedin et al. [8] (2019)	Not specified	20	RF, MLP, J48	PDD	98.11% / –
Mat Rani et al. [9] (2023)	TreeSHAP, Info Gain	15	XGBoost, RF, NB	PDD	97.8% / –
Our Work	RFE-CV	23	CNN	PDD	**97.50% / 95.50%**
Our Work	RFE-CV	23	LightGBM	PDD	**98.90% / 98.90%**
Our Work	RFE-CV	23	MLP	PDD	**96.80% / 9680%**

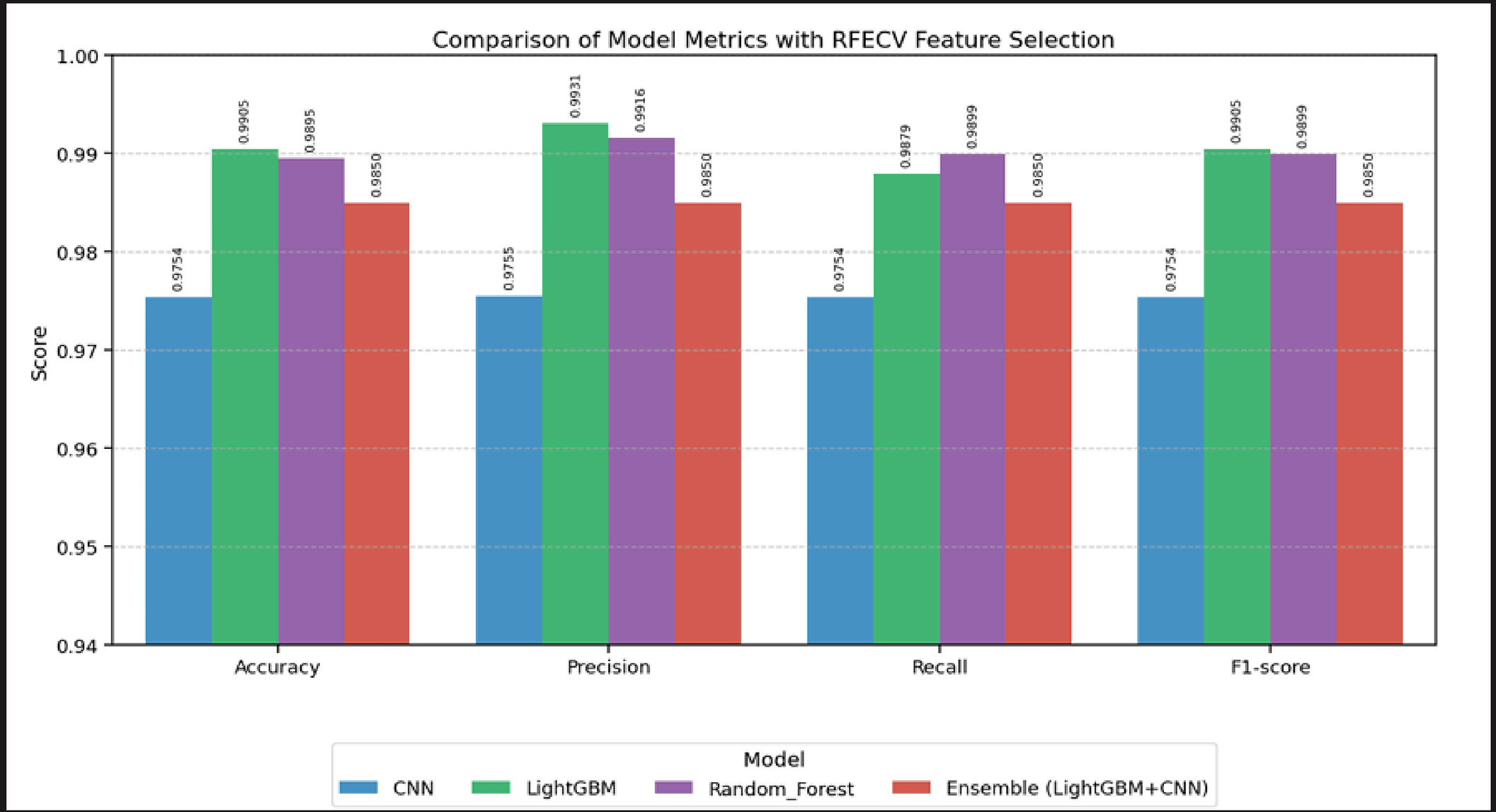
EXPLORING NEW FRONTIERS: PHISHFOE

BALANCING TECHNIQUES COMPARISON

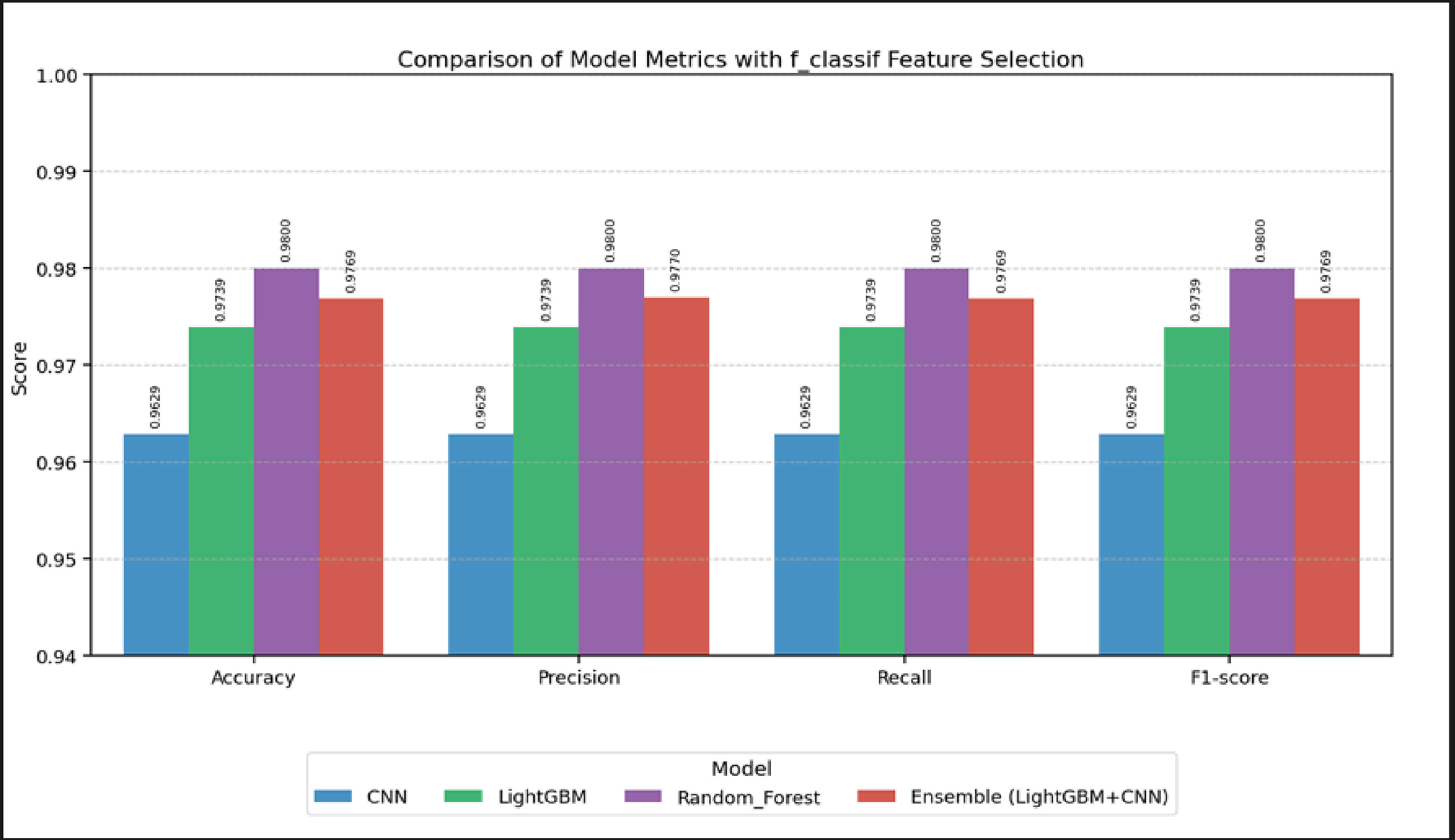
Table 3.1: Performance Metrics for Different Algorithms and Sampling Techniques

Sampling	Algorithm	Metrics
None	Random Forest	*Acc: 0.9750, Prec: 0.9815, Rec: 0.9750, F1: 0.9750*
	LightGBM	***Acc: 0.9885, Prec: 0.9902, Rec: 0.9885, F1: 0.9885***
	CNN	*Acc: 0.9650, Prec: 0.9652, Rec: 0.9650, F1: 0.9650*
	LightGBM+CNN	*Acc: 0.9792, Prec: 0.9850, Rec: 0.9792, F1: 0.9792*
SMOTE	Random Forest	*Acc: 0.9805, Prec: 0.9916, Rec: 0.9899, F1: 0.9899*
	LightGBM	***Acc: 0.9905, Prec: 0.9931, Rec: 0.9879, F1: 0.9905***
	CNN	*Acc: 0.9754, Prec: 0.9755, Rec: 0.9754, F1: 0.9754*
	LightGBM+CNN	*Acc: 0.9850, Prec: 0.9850, Rec: 0.9850, F1: 0.9850*
Undersampling	Random Forest	*Acc: 0.9789, Prec: 0.9805, Rec: 0.9789, F1: 0.9789*
	LightGBM	***Acc: 0.9886, Prec: 0.9905, Rec: 0.9916, F1: 0.9886***
	CNN	*Acc: 0.9675, Prec: 0.9671, Rec: 0.9675, F1: 0.9675*
	LightGBM+CNN	*Acc: 0.9850, Prec: 0.9851, Rec: 0.9850, F1: 0.9850*

FEATURE SELECTION METHODS



FEATURE SELECTION METHODS



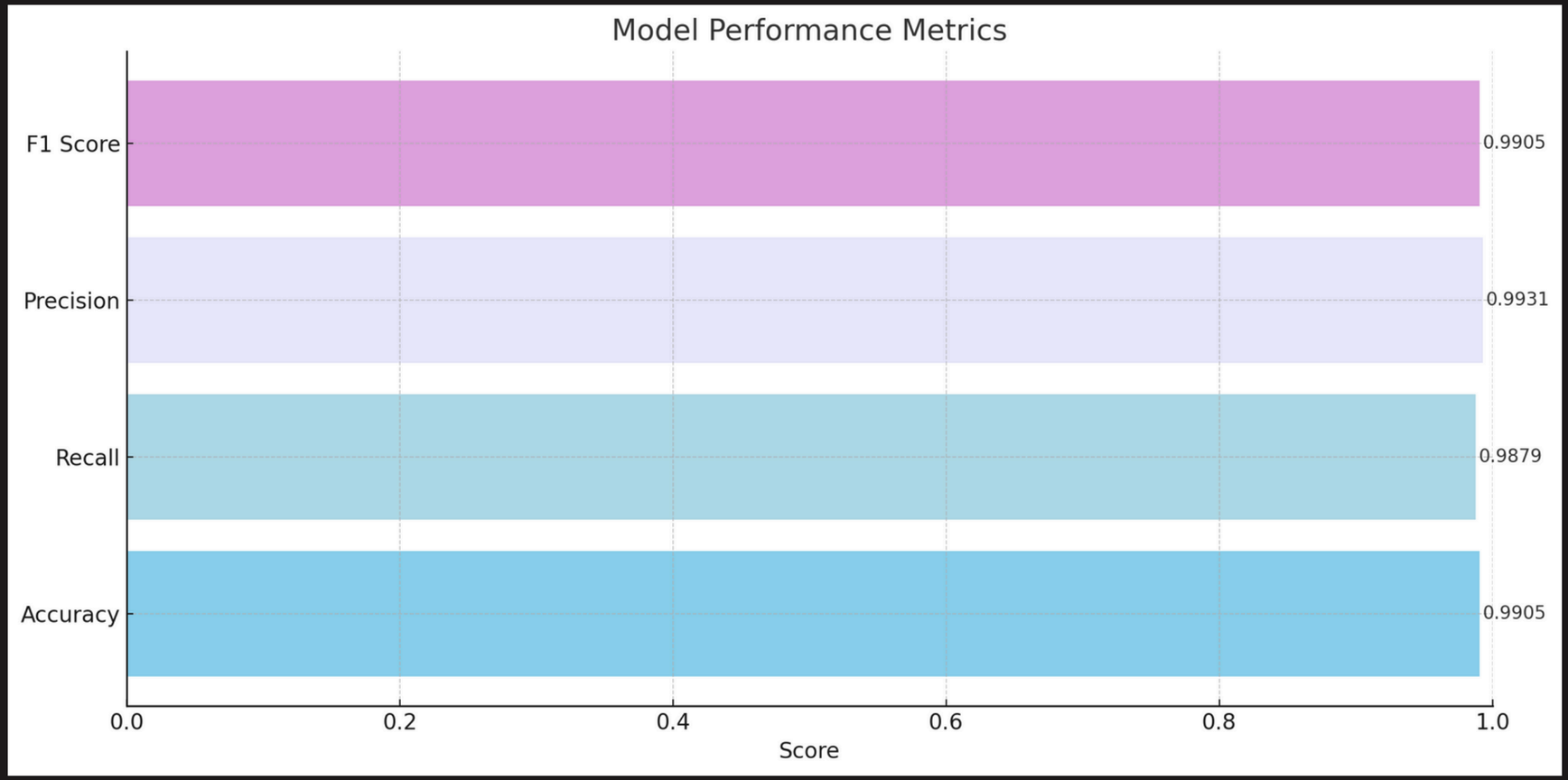
FEATURE SELECTION METHODS

RFE-CV VS F-CLASSIF COMPARISON (F1-SCORE):

Model	RFE-CV F1-Score	f-classif F1-Score
CNN	97.54%	96.29%
LightGBM	99.05%	97.39%
Random Forest	98.99%	98.00%
LightGBM + CNN	98.50%	97.69%

THE BEST RESULT

- The best results were achieved using the **LightGBM** model.
- **Synthetic Minority Over-sampling Technique(SMOTE)** weas used for balancing.
- **Recursive Feature Elimination with Cross-Validation (RFE-CV)** was used for feature selection.
- This setup was applied to the **PhishFOE dataset**.



ADDITIONAL MODELS EVALUATED

OTHER MODELS TESTED:

Model	Accuracy	F1-Score	Precision	Recall
XGBoost	98.54%	98.53%	98.84%	98.22%
XGBoost + LSTM	98.91%	98.91%	99.00%	98.82%
LSTM	97.69%	97.64%	97.66%	97.75%
Roberta Transformer	94.05%	93.81%	96.44%	91.31%
MLP	95.36%	95.36%	94.82%	94.25%



THINK BEFORE YOU CLICK!

DEPLOYMENT

When in doubt, throw the link out!

SYSTEM ARCHITECTURE

- 🔧 Backend: Python-powered API with a trained ML model
- 🌐 Frontend: Web dashboard for real-time interaction
- 🔄 Data Flow:

    1. User submits a URL
    2. Features are extracted (length, HTTPS, suspicious keywords...)
    3. Data is preprocessed
    4. URL analyzed by a classification model
    5. Legitimacy score calculated
    6. Results displayed visually

ANALYSIS ENGINE

URL FEATURE EXTRACTION

- Protocol type, domain length, number of special characters
- Risk indicators: recent domain, shortened URLs, suspicious keywords
- Domain metadata from WHOIS: creation date, registrar information

HTML CONTENT INSPECTION

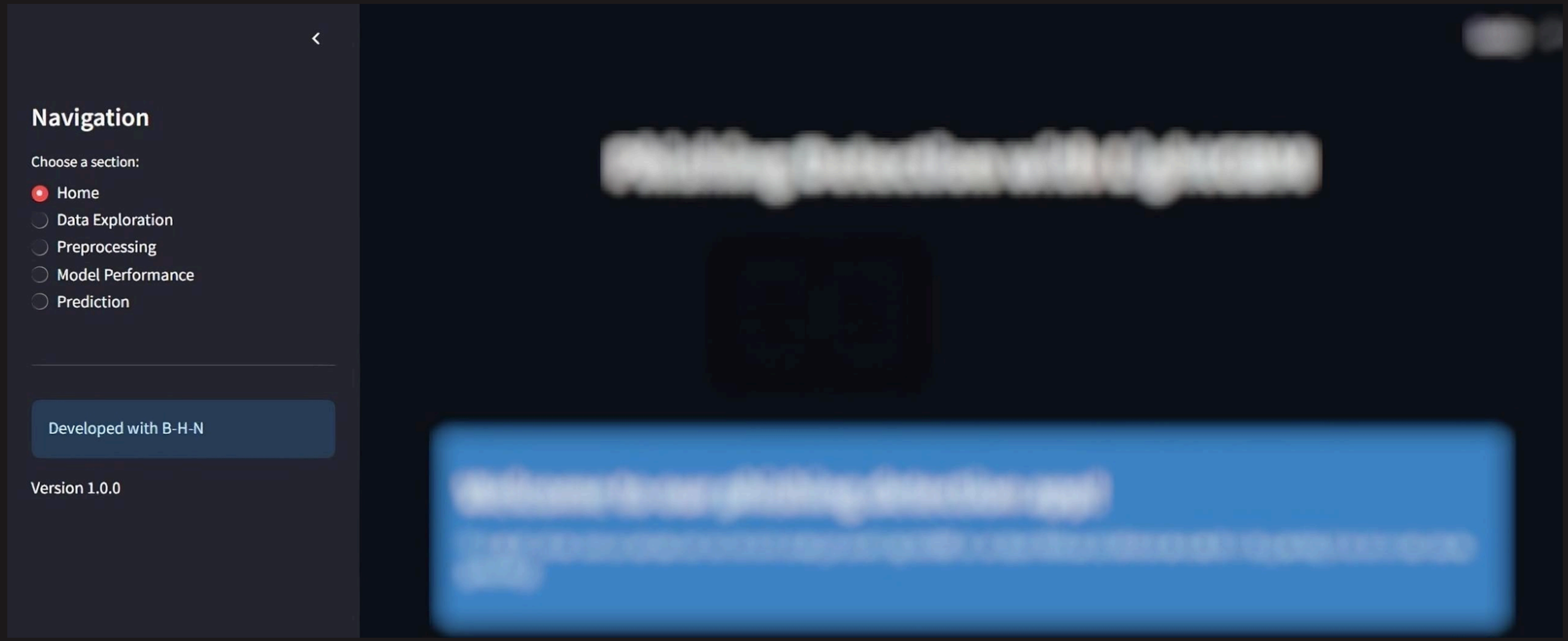
- Presence of login forms, iframes, redirects
- Visual cues: favicon, meta tags, copyright
- Legitimate indicators: social media links, content structure

ML INFERENCE (LIGHTGBM)

- Processed features passed to the model
- Output: Binary classification + legitimacy score (0–100%)

USER INTERFACE

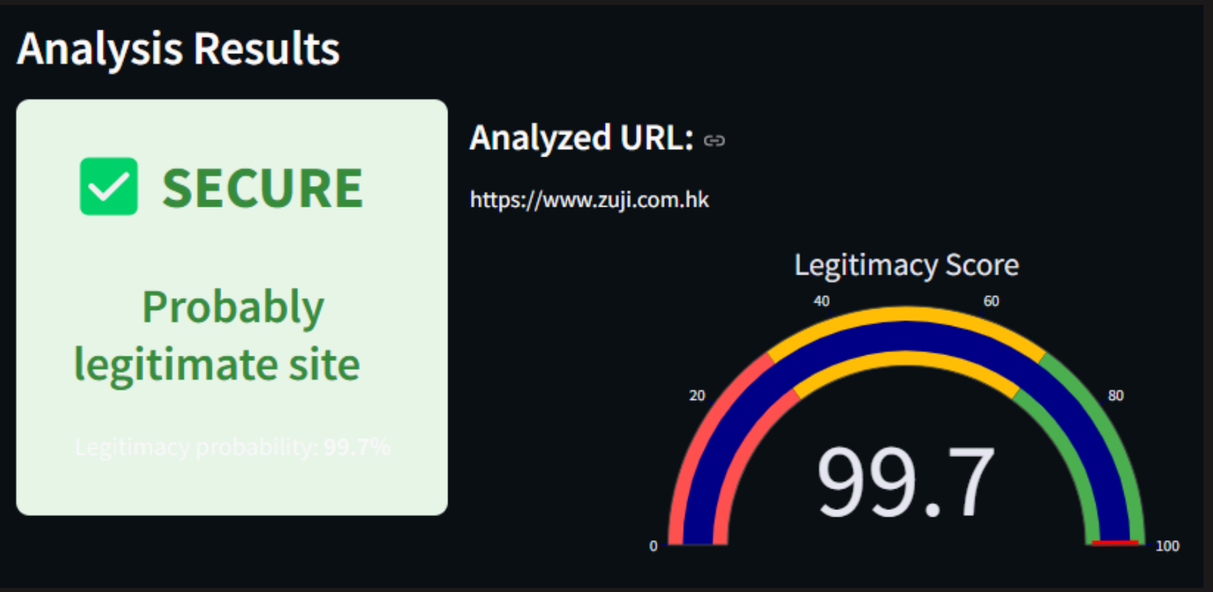
Interactive Dashboard: Section-based layout

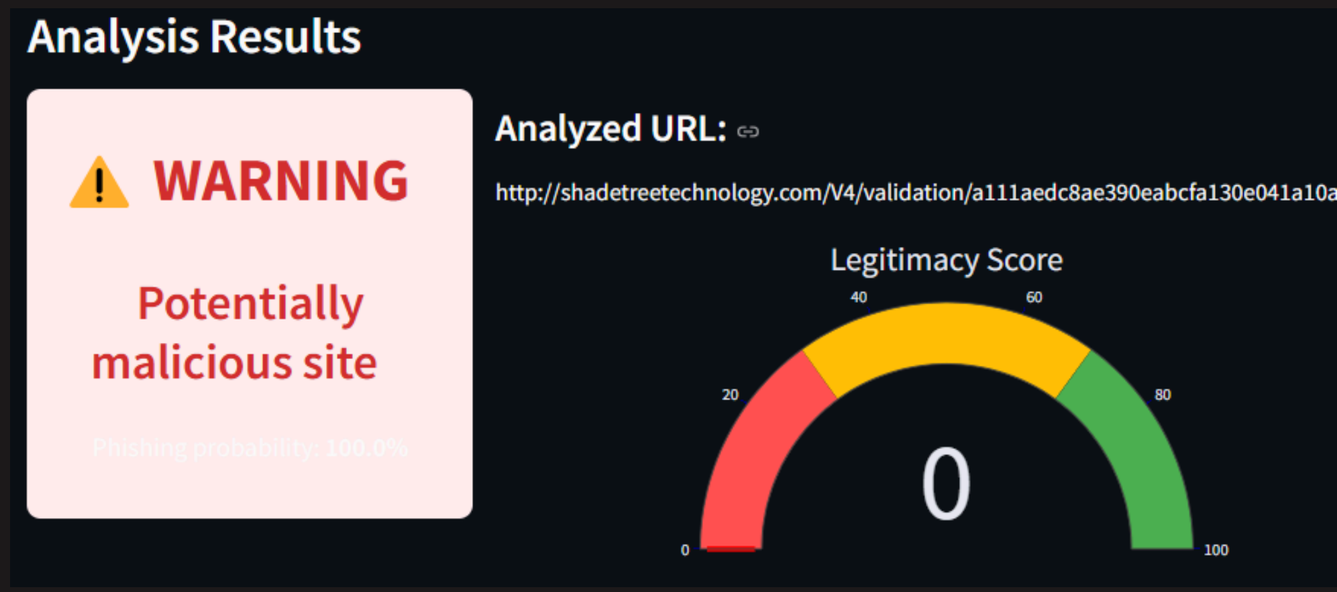


USER INTERFACE

Prediction View:

- Color-coded output: 🟢 Legitimate / 🔴 Malicious
- Confidence score with gauge meter





USER INTERFACE

Detailed Insights:

- List of features with impact on prediction
- Explanation for the risk score



SYSTEM CHARACTERISTICS



PERFORMANCE

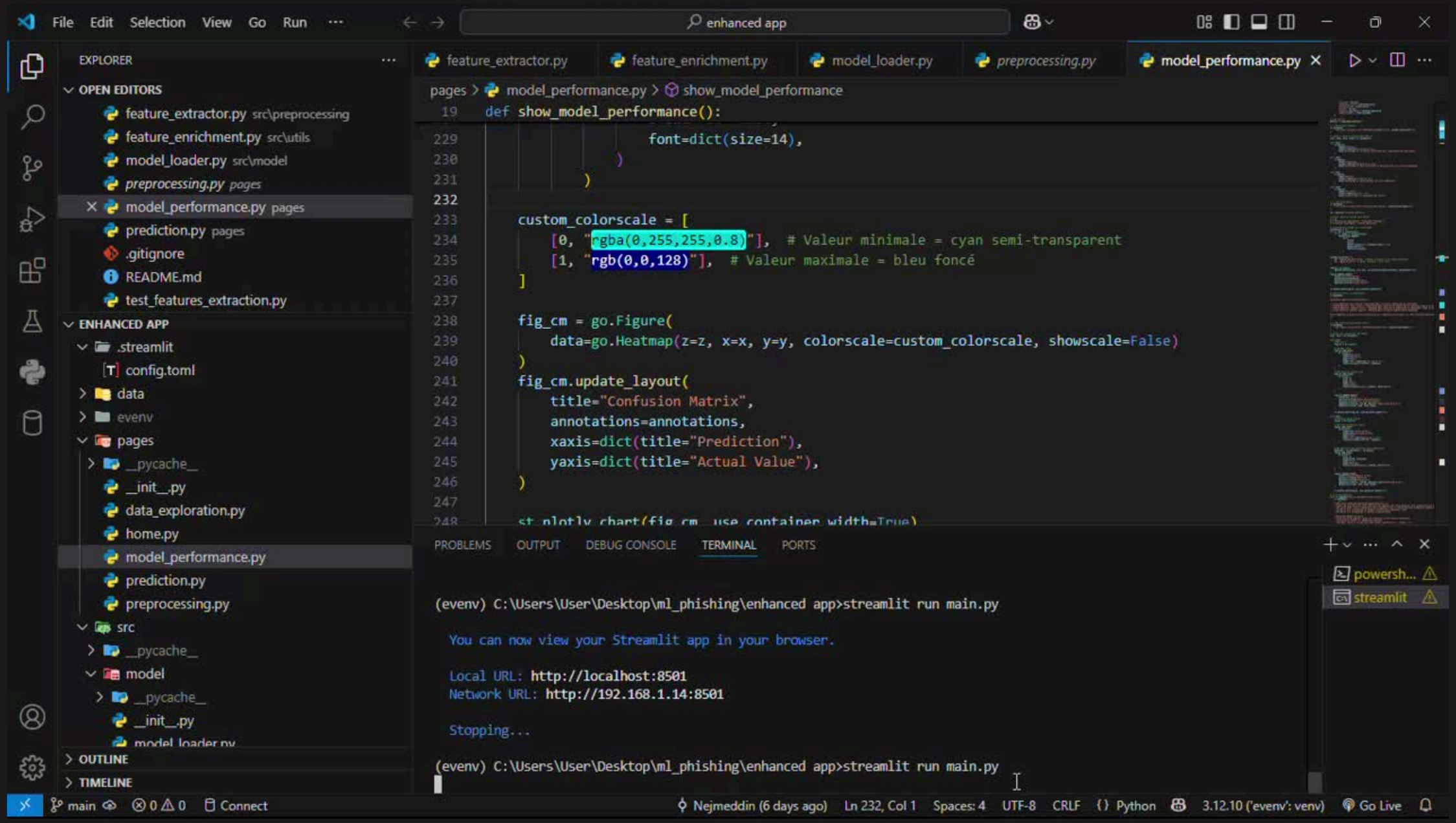
- Optimized pipeline with retry mechanism
- Efficient preprocessing and prediction flow
- Error handling with fallback values



USABILITY

- Responsive UI for mobile/desktop
- Real-time feedback during analysis
- Safety tips and contextual recommendations

DEMONSTRATION





THINK BEFORE YOU CLICK!

CONCLUSION

Links in disguise lead to surprise!

IN CONCLUSION :

- The project delivers a robust phishing detection system using advanced ML and DL techniques.
- LightGBM achieve highest metrics with proper preprocessing and feature selection.
- A user-friendly, scalable dashboard enhances accessibility and real-time detection.



Remember : No system is highly resilient — so always, think twice before you click!



THANKS FOR YOUR ATTENTION !

If you have any questions
Please do not hesitate to ask
We will be readily available to assist you.